

Complexity Reduction of SD-FEC by Bayes-Based Simple-Soft-Output MLSE in Coherent Transmission with Chase Decoder

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Abstract We evaluate the applicability of Bayes-based simple-soft-output MLSE to coherent transmission systems with LR-FEC and turbo product codes. Experimental results show that the proposed scheme reduces the number of test patterns and the number of iterations in Chase decoding by approximately 50% and 25%, respectively. ©2026 The Author(s)

Introduction

As cloud services driven by generative AI and remote work continue to proliferate, communication traffic has been increasing rapidly. To support these services, massive volumes of traffic are exchanged within data centres, creating a growing demand for more cost-effective, high-capacity optical transmission technologies for short-reach applications. Digital coherent optical transmission systems can realize high-capacity data transmission by leveraging advanced signal processing techniques, and they have been employed not only in long-haul optical transmission systems but also in short-reach links between data centres. Transmission capacities exceeding 1.6 Tb/s will soon be required. However, as symbol rates continue to increase, bandwidth limitations imposed by the optical and electrical devices constituting optical transceivers may become a major source of signal degradation. In addition, the degradation in receiver sensitivity associated with higher transmission speeds also poses a significant challenge.

To address these issues, the application of maximum likelihood sequence estimation (MLSE), which provides high demodulation performance even under severe bandwidth constraints, has been investigated [1–8]. Furthermore, soft-decision forward error correction (SD-FEC), which offers high noise tolerance, has been adopted as a standard scheme by the OIF and IEEE. In particular, in coherent optical transmission systems, from the viewpoints of decoding performance and implementation feasibility, LR-FEC [9, 10], which achieves high error-correction performance by applying soft-decision decoding (SDD) based on Chase decoding to BCH codes, and OFEC [11, 12], which further improves error-correction capability by iteratively applying Chase decoding to turbo product codes (TPCs) composed of BCH component codes, have been standardized. Therefore, SD-FEC schemes employing Chase decoding have attracted significant attention.

Chase decoding requires information on the

log-likelihood ratios (LLRs) derived from the received signal during the decoding process [13]. For this reason, a method for simple LLR computation for signals demodulated by MLSE has been proposed [1]. Although this method is a simple-soft-output (SSO) MLSE scheme capable of computing LLRs with computational complexity equivalent to that of the hard-decision Viterbi algorithm (HD-VA), it is based on a heuristic approach and therefore cannot accurately reflect the information obtained from the HD-VA in the resulting LLR values. In contrast, we have proposed an LLR calculation method based on Bayesian updating and demonstrated its applicability to PAM signal transmission under bandwidth-limited conditions [14]. This method is called Bayes-based simple-soft-output MLSE (BB-SSO-MLSE). In this paper, we show that, by applying BB-SSO-MLSE to 16 quadrature amplitude modulation (QAM) coherent transmission, more accurate LLRs can be obtained with computational complexity equivalent to that of the conventional SSO-MLSE scheme. We also demonstrate that the proposed method can reduce the computational complexity associated with SDD for LR-FEC and TPCs.

Principle of BB-SSO-MLSE

Equation (1) shows Bayes' theorem with three variables. BB-SSO-MLSE is based on the theorem, and Eq. (2) is derived from it [14].

$$P(X|Y, Z) = \frac{P(X|Y)P(Z|X, Y)}{P(Z|Y)}, \quad (1)$$

$$\lambda = \lambda' - (-1)^b \left(\ln \frac{1-p}{p} - m \right), \quad (2)$$

$$m \triangleq i_{b=1}(\hat{b}, (-1)^b \lambda') - i_{b=0}(\hat{b}, (-1)^b \lambda'), \quad (3)$$

$$i_x(Z, Y) \triangleq \ln \frac{P(Z, Y|X)}{P(Z|X)P(Y|X)}, \quad (4)$$

where λ denotes the LLR obtained by BB-SSO-

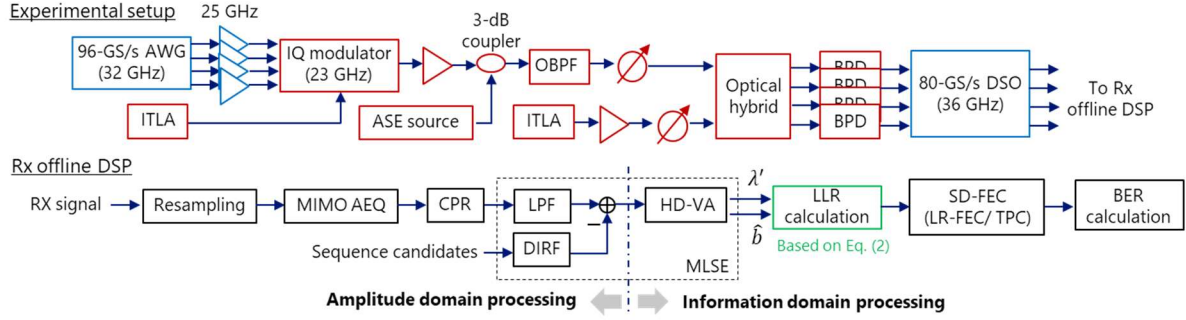


Fig. 1: Experimental setup and DSP block diagram.

MLSE, while λ' denotes the LLR computed from the node metrics obtained by forward sequence estimation in the HD-VA [1, 15], and $\lambda' \in Y$. $\lambda' < 0$ when $b = 0$ is more likely. b is transmitted bit, and $b \in X \triangleq \{0,1\}$. $\hat{b}$ denotes the HD output of the HD-VA, and $\hat{b} \in Z \triangleq \{0,1\}$. The parameter p represents the bit error rate (BER) of $\hat{b}$ and serves as an indicator of its reliability. The parameter m denotes the difference in conditional pointwise mutual information (PMI) between the cases $b = 0$ and $b = 1$, and thus serves as a measure of the correlation between λ' and $\hat{b}$. If these are completely independent, then $m = 0$. The information used by BB-SSO-MLSE to calculate the LLR is only the information obtained by the HD-VA, which is the same information used by the conventional SSO-MLSE [1]. Equation (2) is applicable to probabilistically-constellation-shaped (PCS) signals, and $P(X = 0) \neq P(X = 1)$ in λ' is used for PCS signals.

Experiment and Results

We evaluated the applicability of BB-SSO-MLSE to coherent optical transmission and its impact on the decoding performance of LR-FEC and TPCs. Figure 1 shows the experimental setup. A polarization-division-multiplexed (PDM) 16QAM optical signal is generated using an arbitrary waveform generator (AWG), a driver amplifier, and a PDM-IQ modulator. The sampling rate of the AWG is 96 GS/s, and the signal baud rate is 64 GBd. The electrical waveform is shaped by a Nyquist filter with a roll-off factor of 0.1. The wavelength of the integrable tuneable laser assembly (ITLA) is set to 1555.344 nm. The optical signal is subject to severe bandwidth limitations imposed by the AWG, modulator, and driver amplifier. Amplified spontaneous emission (ASE) noise is added through a 3-dB coupler. After out-of-band ASE is removed by a 100-GHz optical band-pass filter (OBPF), the signal is combined with a local oscillator (LO) of which power is 14.5 dBm. The received optical signal-to-noise ratio (OSNR) is set to 28 dB. The optical signal is converted into an electrical signal by balanced photodetectors (BPDs), and it is sampled at 80 GS/s by a digital storage oscilloscope (DSO).

Figure 1 also shows the block diagram of the receiver-side digital signal processing (DSP). After resampling, the signal is equalized by a multiple-input and multiple-output adaptive equalizer (MIMO-AEQ) [16], followed by a carrier phase recovery (CPR). MLSE is independently applied to each of the four lanes IX, IY, QX, and QY. The low-pass filter (LPF) and desired-impulse-response filter (DIRF) used in the MLSE both have three taps, and the traceback length in the HD-VA is 15. Then, λ' and $\hat{b}$ are obtained. The LLR values are calculated from Eq. (2). m is treated as a hyperparameter in this evaluation.

Figure 2(a) shows the relationship between m in Eq. (2) and the post-FEC BER for LR-FEC. The post-FEC BER is obtained by Chase-2 decoding applied to the BCH(126,110) code used as an inner code in LR-FEC. In the decoding process, the sign of the bitwise LLR is inverted when $b = 1$. This means that the calculation of post-FEC BER regards all transmitted bits as 0 [17, 18]. For each plot, approximately 8 million bits are used to calculate the post-FEC BER. The figure shows several cases with a different number of test patterns (TPs) generated in Chase-2 decoding. As shown in Fig. 2(a), the post-FEC BER decreases as the number of TPs increases. It is also confirmed that increasing the number of TPs beyond 64 does not significantly improve the error-correction performance. In addition, the value of m that minimizes the post-FEC BER is found to be approximately $m = 2$ regardless of the number of TPs.

Figures 2(b) and 2(c) show the relationship between m and the post-FEC BER for a TPC with BCH(256,247) component codes. The post-FEC BER is obtained by Chase-Pyndiah (CP) decoding with three and four iterations applied to the TPC. The α and β parameters used in CP decoding are set to commonly used values [19]. For each plot, approximately 18 million bits are used to calculate the post-FEC BER. As shown in Figs. 2(b) and 2(c), the post-FEC BER decreases as the number of TPs increases. The value of m that minimizes the post-FEC BER is found to be approximately $m = 2$, and the optimal value is independent of both the number of TPs and the number of iterations. This value is almost the same as

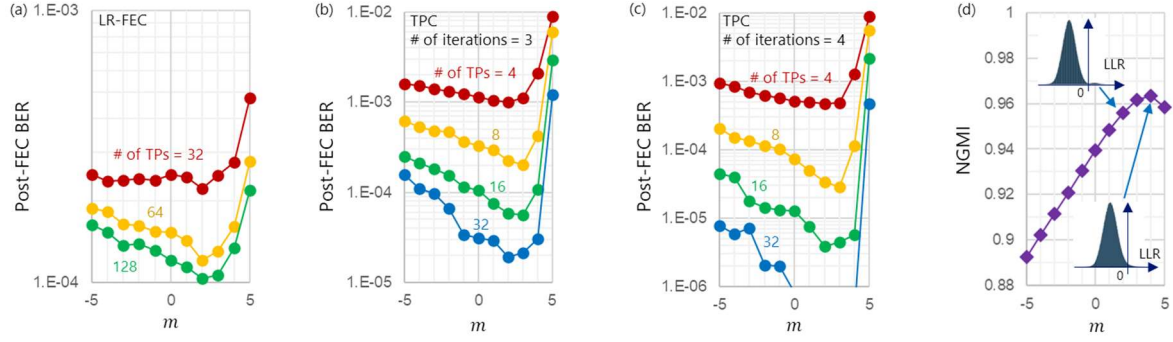


Fig. 2: Relationship between m and (a) post-FEC BER for LR-FEC, (b) (c) post-FEC BER for TPCs, (d) NGMI.

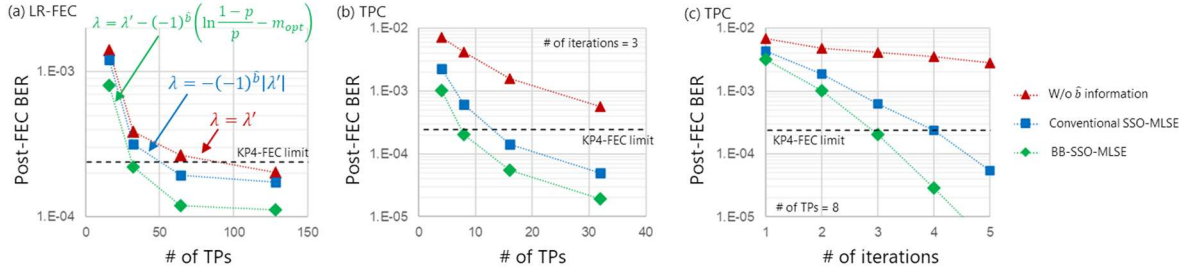


Fig. 3: Relationship between (a) (b) $\#$ of TPs and post-FEC BER, (c) $\#$ of iterations and post-FEC BER.

the optimal value for LR-FEC. On the other hand, compared with the case of Chase-2 decoding, the post-FEC BER in CP decoding is more sensitive to the value of m . This is considered to result from the iterative operations of CP decoding. This result implies that the more accurate the initial LLR values are, the more accurately they can be refined through the iterative operations in CP decoding.

Figure 2(d) shows the relationship between m and the normalized generalized mutual information (NGMI) [20, 21], where the scale of LLR is optimized for λ' . The figure also shows histograms about $(-1)^b \lambda$ for the most significant bits. As shown in the figure, the maximum NGMI is achieved at $m = 4$. This result indicates that the value of m that maximizes the NGMI does not necessarily minimize the post-FEC BER. This may be because the LLR distribution generated by Eq. (2) deviates from a distribution derived from additive white Gaussian noise (AWGN).

Figure 3(a) shows the relationship between the number of TPs and the post-FEC BER for LR-FEC in three cases: when the information of $\hat{\delta}$ is not used in the LLR calculation, when LLRs are calculated using the conventional SSO-MLSE [1], and when LLRs are calculated using BB-SSO-MLSE based on Eq. (2). We use the optimal value of m that minimizes the post-FEC BER. The dashed line in the figure indicates the threshold of KP4-FEC used as the outer code of LR-FEC. As shown in the figure, by applying the proposed BB-SSO-MLSE, the number of TPs required to achieve the same post-FEC BER can be reduced by half compared with the

conventional scheme. In Chase decoding, including CP decoding, BCH hard-decision decoding (HDD) must be performed for all TPs, and this BCH decoding process is the dominant contributor to the computational complexity of Chase decoding. Therefore, reducing the number of TPs directly contributes to a reduction in computational complexity.

Figure 3(b) shows the results for TPC with BCH(256,247) component codes. As shown in the figure, in CP decoding as well, applying the proposed BB-SSO-MLSE makes it possible to reduce by approximately half the number of TPs required to achieve the same post-FEC BER compared with the conventional SSO-MLSE. Figure 3(c) shows the relationship between the number of iterations and the post-FEC BER for the TPC. The number of TPs is fixed at 8. As shown in the figure, by applying the proposed BB-SSO-MLSE, the number of iterations required to achieve the same post-FEC BER can be reduced by approximately 25%. Since the number of iterations in CP decoding is a dominant factor not only in computational complexity but also in processing latency, this reduction in the number of iterations also contributes to latency reduction.

Conclusion

We investigated the applicability of BB-SSO-MLSE to coherent transmission with a Chase decoder. We showed that BB-SSO-MLSE reduces the computation complexity of LR-FEC and TPC decoding, where the number of TPs is reduced by approximately 50% and the number of iterations is reduced by approximately 25%.

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